Rohan Shrivastava | 732-841-1484 | srohanshri@gmail.com | linkedin.com/in/rohan-shrivastava456

EDUCATION

Rutgers University – New Brunswick, NJ
B.S. in Electrical and Computer Engineering — May 2028

TECHNICAL SKILLS

- **Programming**: Java, MATLAB, C++, C, Python, HTML
- **Embedded & Hardware:** Embedded C++, Arduino, ESP32(C3,S3), I2C/SPI Protocol, RP2040, RP2350
- **Tools & Software:** KiCad, EasyEDA, Arduino IDE, Onshape, SolidWorks, Git/Github, VS Code

PROFESSIONAL EXPERIENCE

Automation QA & AI Systems Intern | *Accelirate* | May 2026–July 2026

- Engineered test suites for an **Agentic framework-driven QA application** (Accel QA), using **custom SQL datasets** to simulate edge-case scenarios and accelerate testing cycles.
- Isolated and debugged application-level issues across **backend services**, improving **system reliability** and **fault tolerance**.

TECHNICAL PROJECTS

Breathe.exe | *HackRU Fall 2026 Best use of Presage Winners* | Oct 10–11, 2026

- Co-created a horror-themed interview simulation game in 24 hours using **Unity** and a **Gemini-based agent t**hat drives the interviewer's actions and responses from live biometric stress data.
- Integrated **ElevenLabs** for dynamic voice generation and **vocal stress analysis**, plus **Presage camera tracking** for facial **stress patterns** and **heart rate**.
- Designed a custom **glove-based input device** to make the game more immersive, utilizing a **6-axis IMU** and joystick inputs powered by an **ESP32-C3** transmitting data through **ESP-NOW (10 ms latency)** to an **ESP32 receiver**.

Soil Node | July 2026–Present

- Developing a small **IoT** plant monitor powered by an **ESP32-C3** microcontroller that utilizes **BLE 5.0 (40 ms latency)** to send temperature, humidity, and soil moisture telemetry to a **Python logging application** that processes sensor data and **generates real-time charts and automated plant health recommendations**.
- Duty-cycled the sensors to sample temperature, humidity, and soil moisture for 5 seconds each hour and log battery level hourly, for an estimated **3-month battery life**.

IAM3D Project Lead & Treasurer | *ASME* | Sept 2026–Present

- Directed engineering team through the design, rapid prototyping, and additive manufacturing of custom robotic systems for the IAM3D competition.
- Managed **organizational budgets**, allocated funds for hardware procurement and prototyping materials, and oversaw financial operations for competition projects.

Electrical Co-Lead | *RU HART* | Apr 2025–Present

- Designed a 4-layer **RP2040-based motor driver (20A, 12V**) with **thermal shutdown**, **overcurrent and undervoltage protection**, and current and temperature monitoring, communicating with an **NVIDIA Jetson Nano** over **CAN 2.1.**
- Built an **ESP32-S3** mapping robot with a **5 Hz LiDAR** that scans an enclosed space and streams the geometry to a **local web server** over **Wi-Fi** in real time.
- Trained **20+ members** in **KiCad Schematic & PCB design**, electrical fundamentals, and system design.
- Co-organized and led multi-week robot workshops, mentoring members through hands-on development of small robots and **embedded systems-based projects**.